Activity 7.2: Surface Area & Volume

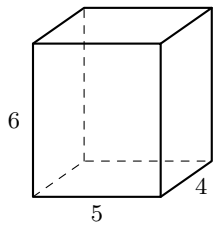
A

Name: _____ Partner's name: _____ Date: _____

Directions: Find the volume or surface area asked for. Give exact answers in terms of π when needed. Diagrams are accurate, but use math rather than a ruler or protractor.

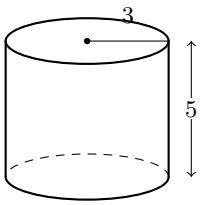
Work alone, then compare after each problem. If answers differ, find the mistake together.

1 Find the volume of the rectangular prism.



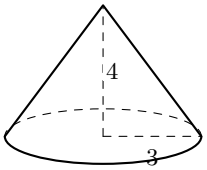
Answer: _____ Match? ☐ Yes ☐ No

2 Find the volume of the cylinder.



Answer: _____ Match? ☐ Yes ☐ No

3 Find the volume of the cone.



Answer: _____ Match? ☐ Yes ☐ No

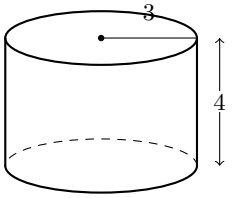
4 Find the volume of a sphere with radius 3.

Answer: _____ Match? ☐ Yes ☐ No

5 Find the surface area of a cube with edge length 5.

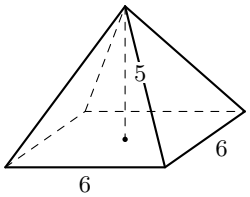
Answer: _____ Match? ☐ Yes ☐ No

6 Find the total surface area of the cylinder.



Answer: _____ Match? ☐ Yes ☐ No

7 Find the volume of the square pyramid.



Answer: _____ Match? ☐ Yes ☐ No

8 Two similar solids have a scale factor of 2 : 3. The smaller solid has volume 24. Find the volume of the larger solid.

Answer: _____ Match? ☐ Yes ☐ No

Activity 7.2: Surface Area & Volume

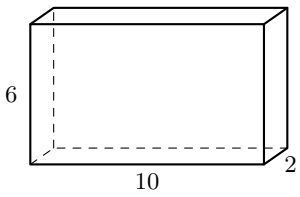
B

Name: _____ Partner's name: _____ Date: _____

Directions: Find the volume or surface area asked for. Give exact answers in terms of π when needed. Diagrams are accurate, but use math rather than a ruler or protractor.

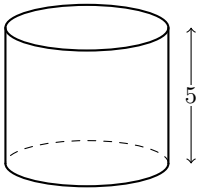
Work alone, then compare after each problem. If answers differ, find the mistake together.

1 Find the volume of the rectangular prism.



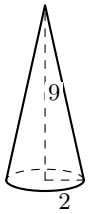
Answer: _____ Match? ☐ Yes ☐ No

2 A cylinder has base area 9π and height 5. Find its volume.



Answer: _____ Match? ☐ Yes ☐ No

3 Find the volume of the cone.



Answer: _____ Match? ☐ Yes ☐ No

4 A sphere has surface area 36π. Find its volume.

Answer: _____ Match? ☐ Yes ☐ No

5 Find the surface area of a cube with volume 125.

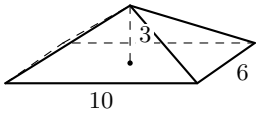
Answer: _____ Match? ☐ Yes ☐ No

6 Find the total surface area of the cylinder.



Answer: _____ Match? ☐ Yes ☐ No

7 Find the volume of the rectangular pyramid.



Answer: _____ Match? ☐ Yes ☐ No

8 Two similar solids have side lengths in the ratio 1 : 3. The smaller solid has volume 3. Find the volume of the larger solid.

Answer: _____ Match? ☐ Yes ☐ No